Answer key - Solving Linear Equations of the First Degree

GCE Ordinary Level - Mathematics - Corrige

Detailed method for every exercise. Follow the substitutions line by line.
Corrige detaille de chaque exercice. Suis les substitutions ligne par ligne.

1. Solve 3x + 5 = 14.
Résous 3x + 5 = 14.

- x = 3 *
- x = 9
- x = 19
- x = 9/5

Answer: x = 3
Subtract 5 from both sides: 3x = 9. Divide by 3: x = 3.
On soustrait 5 de chaque membre : 3x = 9. On divise par 3 : x = 3.

2. Solve 2x - 7 = x + 3.
Résous 2x - 7 = x + 3.

- x = 10 *
- x = 4
- x = -4
- x = 5

Answer: x = 10
Subtract x from both sides: x - 7 = 3. Then add 7: x = 10. Check: 2(10) - 7 = 13 and 10 + 3 = 13.
On soustrait x aux deux membres : x - 7 = 3. Puis on ajoute 7 : x = 10. Vérification : 2(10) - 7 = 13 et 10 + 3 = 13.

3. Solve x/4 - 2 = 1. Give the value of x.
Résous x/4 - 2 = 1. Donne la valeur de x.

Answer: 12
Add 2 to both sides: x/4 = 3. Multiply both sides by 4: x = 12.
On ajoute 2 aux deux membres : x/4 = 3. On multiplie les deux membres par 4 : x = 12.

4. Solve 5x = 3x + 8.
Résous 5x = 3x + 8.

- x = 4 *
- x = 8
- x = 2
- x = 16

Answer: x = 4
Subtract 3x from both sides: 2x = 8. Divide by 2: x = 4. Check: 5(4) = 20 and 3(4) + 8 = 20.
On soustrait 3x aux deux membres : 2x = 8. On divise par 2 : x = 4. Vérification : 5(4) = 20 et 3(4) + 8 = 20.

5. What happens with the equation x + 4 = x?
Que se passe-t-il avec l'équation x + 4 = x ?

- It has no solution / Elle est sans solution *
- It has the solution x = 4 / Elle a la solution x = 4
- It has the solution x = -4 / Elle a la solution x = -4
- It has infinitely many solutions / Elle a une infinité de solutions

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Answer: It has no solution

Subtracting x from both sides leaves 4 = 0, which is false. The equation has no solution.

En soustrayant x aux deux membres, il reste 4 = 0, ce qui est faux. L'équation est sans solution.

6. Solve 7x + 3 = 5x + 11. Give the value of x.

Résous 7x + 3 = 5x + 11. Donne la valeur de x.

Answer: 4

Subtract 5x from both sides: 2x + 3 = 11. Subtract 3: 2x = 8. Divide by 2: x = 4.

On soustrait 5x aux deux membres : 2x + 3 = 11. On soustrait 3 : 2x = 8. On divise par 2 : x = 4.

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